

VoIP Protocols

Module 45

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hello viewers I welcome you to the lecture series on cyber security information security today we will discuss about V IP protocol that is voice over IP protocol so to start with a question arises that what is viip we know that voice can be digitalized the digitalized voice can be transmitted in packets over the network Voice transmission comes in the different flavors in the past t communication through fixed circuit switch network has dominated in recent years we see data communication Network especially voice over IP if you have never heard of VI IP get ready to change the way you think about longdistance phone calls viip or voice over Internet Protocol is a method for taking analog audio signals like the kind you hear when you talk on the phone and turning them into the Digital Data that can be

transmitted over the Internet voiceover Internet Protocol is the methodology and group of Technologies for the delivery of voice Communications and multimedia sessions over Internet Protocol networks such as Internet Voiceover IP is the transmission of voice traffic in packet using the internet as the transmission medium IP is used rather than the traditional circuit transmission viip is defined in the recommendations it u t h. 32x and RFC 2443 why viip the Internet Protocol is used to deliver packets carrying digitalized voice however IP was not designed for realtime traffic such as voice and video communication IP is the connectionless protoc call meaning a virtual connection is not established through a network prior to

transmission IP makes no guarantees concerning reliability flow control error detection or error correction potential errors include out of sequence packets or even loss of packets Voice transmission requires guaranteed connection and a reasonable delay nevertheless IP succeeds partially due to the high cost Associated we the traditional circuit switch TDM Network viip uses packet switch Network it makes the network transparent to the upper layers that are involved in Voice transmission through an IP based Network the existing use of IP network also allows the integration of voice and data integration to leverage the connectionless nature of Ip vendors have developed higher layer protocols to address the guaranteed connection and

3 minutes, 1 second transmission issue viip and OSI model viip follows a layered model comparable to the OSI 7 layered model we are familiar with just like the purpose of developing The OSI model breaking into defined layers provides a framework it establishes a standard to

make the system more manageable and flexible each layer is relatively independent to other layers around it changes made to one layer should have no or minimal impact on other layers what are the V protocols the two most generally used protocols for Vi area unit the itu normal h.323 and also the iatf normal sip each area unit signaling protocols that discovered maintain and terminate a VI decision Additionally the media entry management protocol mgcp provides a

signaling and management protocol between viip gateways and ancient pstn public switch telephone Network gateways first protocol is itut h.323 h.323 may be a comprehensive protocol underneath the itut specifications for for cation voice video and information across a network the h.323 specification includes many subprotocols h225 for specifying decision controls example decision setup and tear down h235 for specifying the protection framework for h.323 and also the decision setup up h245 for specifying media methods and parameter negotiations like terminal capabilities .450 for specifying supplementary services like decision hold and telephony h235 additionally provides security measures like authentication Integrity

privacy and a few non repudiation support in h.323 Communications it's designed to work seamlessly with different protocols like h245 and h225 a decision setup is secured through transport layer security TLS once established a decision management is initiated so secret writing and media channel data will be negotiated h. 32 two three utilizes RTP that is realtime transport protocol or rtcp that is realtime transport management protocol as its transport protocol that rides on high of UDP secret writing is performed among the RTP packet by thirdparty Hardware or at the network layer authentication underneath h.323 will be either radially symmetrical and ention based or subscription based for radially symmetrical encryption based authentication previous contact between the ACT entities isn't

needed as a result of protocol uses defy Helman key exchange to get a shared secret identity between the two entities with relevancy the h. 235 recommendation a subscription based Authentication needs a previous shared secret identity and their area unit three variations of this first password based with the radially symmetrical secret writing second password based with hashing and third certificate based with signatures second protocol is session initiation protocol that is sip sip may be a text based application layer protocol that addresses the signaling and session management among the packet telephone Network it's outlined in RFC 3261 sip uses a request response model in sip authentication and authorization

area unit handled either by employing a lower layer theme or on a request By Request basis with the challenge response mechanism as sip may be a lightweight protocol its security capabilities area unit terribly restricted sip requests and responses can't be end to end encrypted as a result of message feels like the request and root ought to be visible to proxy servers that area unit gift in several Network architectures to make sure sip request area unit rooted properly voice information is transmitted in clear text over UDP and Communications protocol though sip supports s mbased secret writing exploitation digital certificates sure

header fields employed in requests and responses can't be encrypted the SIP protocol depends on

the transport layer security mechanisms like TLS or IPsec to produce the desired security for the entire message. Third, the protocol is Media Entry Management Protocol (MGM) that is managed by the media management social unit as RFC 3435. It expects that MGM messages can be continuously be carried over Secure Net connections as outlined within the science security design as outlined in RFC 2401. Exploitation either the science authentication Header outlined in RFC 242 or the science encapsulating security payload outlined in RFC 246. This enables for information connectionless Integrity origin authentication and no obligatory anti-replay protection of messages passed between the media entry that converts circuit switch traffic to packet

based traffic and also the media entry controller that is MGC that dictates the service logic of the traffic. Next is processing in VIP system at the sender aspect. Analog voice signals are unit reborn into digital signals, compressed and so set into a planned format. Exploitation voice codec like G.711, G.729 and G.723. Etc. Next the encoded voice is softened into equal sized packets. Moreover, in every packet headers from completely different layers are unit hooked up to the encoded voice: the protocol headers, accessor to voice packets, area unit of RTP, UDP and signs further as link layer header. The packet area unit is then sent over the network IP to its destination. Wherever the Desformant and deacetalization of the received packets is disbursed throughout the transmission.

Method inference time variation of packet delivery might occur. Hence a playout buffer is employed that smoothes the playout at the receiver. Finish which can have a delay caused as the result of interference. Packets are unit cued at the playout buffer for a stipulated time before being played. However, packets that arrive later than the playout time are unit discarded. VoIP uses the sign protocol specifically Session Initiation Protocol (SIP) that is SIP and H.323. These sign protocols are unit needed to a certain View new IP calls and to shut the media streams between the purchasers. Another important topic to discuss is quality of service (QoS). In VoIP, QoS is employed for mensuration of the degree of user satisfactions in

different words, quality of service will be outlined because the network ability to produce smart services that satisfy its customers. Once degree of user satisfaction is over, it means that QoS is additionally higher. QoS are unit shortly represented as given below. Delay. Delay will be outlined because the total time it takes since an individual act, another person speak words and hearing them at the opposite finish. Delay will be classified into three categories: delay at the supply, delay at the receiver and network delay. Jitter variation in transmission delay is thought as interference. Science network doesn't guarantee of package delivery time that introduces variations in transmission delay and its additional negative effects on voice quality. Packet loss. Packets transmitted over science

network is also lost within the network or arrived corrupted or late packets would be discarded guarded after they arrive late at the interference buffer of the receiver or once there's overflow in the interference buffer or router buffer therefore packet loss is adequate the full loss happens throughout congestion of network and late arrival throughout the packet loss the sender is aware to conduct the Lost packets it causes additional packet delay and it affects the transmission Qs Eco in VII Eco happens once a caller at the center aspect hears the reflection of his own voice once he talked on the phone or the electroacoustic transducer whereas the coli doesn't notice the Eco Eco is that the term of the reflections of the scent voice signal by the way finish Eco May will be electrical Eco that exists in

pstn networks or Eco of sound that is difficulty in viip networks throughput the output is also outlined because the most range of bits received out of the full range of bits sent throughout associate degree interval of your time next topic we are going to discuss is viip Security First is viip attacks threats attackers typically Target some fashionable and wellknown systems and applications viip has become one in every of such application viip like every another system or applications have its weakness therefore protocol designers ought to address it before with the success putting in viip on a universal scale thus this section presents a study of attacks on the viip Dos denial of service dos attacks that reduces the

quantity of obtainable science addresses information mayor and different router functions a DS attack typically blocks the service of the server a viip based mostly DS attack bombards a decision process application with the massive amounts of coincidental requests that it cannot method inflicting the motility down of the appliance thereby denying service to approved or supposed users DS attacks will be directed toward any network part to disrupt these systems practicality Network sniffing Network sniffing attacks occur once a personal or offender is observant the network traffic patterns typically any system user or attacker on a network that's sharing a transmission medium has the power how to look at different

systems traffic eavesdropping eavesdropping is a shot towards assembling sensitive data to arrange for a Cyber attack or to achieve intelligence in viip E dropping maybe a situation wherever the offender is in a position to watch signal or media contents that area unit changed between users so as to look at communications to arrange for different future attack ta 17 minutes, 1 second spoofing one of the kinds of viip spoofing is display spoofing wherever the offender masqueraders itself as a licensed viip user and places a decision the display that seems on the user aspect seems authentic even supposing the offender is exploitation unfair means that the offender will currently trick the user in giving for free sensitive information spoofing is thus the VIP

version of ancient fishing toall fraud tall fraud is that the ability to possess associate degree unauthorized access to the VIP Services typically for financial gain for VIP suppliers this is often one in every of the for most essential attacks toall FR will be recognized by the manipulating the sign messages or the configuration of VIP Parts together with the charge

systems spam over net telephone that is spit due to its a lot of lower communication prices VIP network has become additional engaging as another to this bstn further as a target for spammers viip spam may be a recorded self- dialed phone calls exploitation foration net protocol VIP spam additionally called spam over net telephone that is spit is popping Dead Set be a significant downside for VIP

networks spit is additional sever that your email spam due of its assaultive nature which needs a period defense reaction what is the security measures for viip according downside on DUS there has been a report that sure viip phones area unit Lael to each do attacks and sure viip routers also are at risk of malicious Traffic Solutions to avoid a DS attack monitoring and filtering to take care of list of suspicious users and deny those users from getting any connection or session authentication to ATT test the identity of a US user before forwarding his or her messages through the network stateless proxy to lower the risks of memory exhaustion du as homeless proxy will be accustomed to perform different Security checks like user authentication third party

registration and filtering spam sources server style to create computer hardware memory and network affiliation the primary line of defense against any dos attacks reported downside on EES dropping an Internet Security Systems team discovered the issues in viip system during a vendor decision manager the vendors offer associate degree offender the power to concentrate in or forward calls additionally to gain unauthorized access to networks running on VI answer to eavesdropping employing perfect Hardware only approved forx or to tend access to wiring closets vulnerable Network purpose or to implement a port based mostly waterproof address security for instance on a reception ctsy phone a method ought to often scan the

network of devices is running in unauthorized mode another answer is secret writing of viip traffic reported downside on spoofing there are many reports that Banks and online payment services were victims of attack wherever the offender referred to as a credit card client and receiv received the purchases into giving for free sensitive data relating to their accounts by declaring that there had been dishonorable activity on their accounts solution to spoofing an effective authentication module combined with secret writing would be an efficient answer to spoofing and masquerading attacks reported downside on tall fraud tall fraud threat is commonly most vital within the eyes of enormous organization an eminent theory of service from from an outsized

organization will go unnoticed for quite your time permitting the offender to rack up an outsized bill at the organization's expense but tall fraud is additionally quite simple to accomplish a straightforward configuration downside within the dial set up with leave lines open for check purpose associate degreed remote exess will be abused by an offender it t but profitable from the purpose of fre of the offender to commit tall fraud as a result of once the attack is over the offender will abuse the service quite speedily by reselling them with no real

prices concerned from the attackers aspect solution to tall fraud viip suppliers will stop tall fraud by configuring powerful Firewalls that stop attacks and by protective the

23 minutes, 1 second ports reported downside on spit it is potential Network for an offender to impersonate as another VIP caller for instance associate degree offender May probably inject a pretend ID into a standard VIP decision so the receiver mistakes the decision to be real and coming back from a best known and short supply the receiver mistaken by the electronic ID of the caller might Place unwarranted trust within the person at the opposite finish in such a communication the receiver is also hawed into revealing personal data like account number SSN or answers to security questions a mother's family name for instance solution to spit a normal blacklisting approach 1 through all the calls coming back from the best known Science address VIP vendors additionally give security

measurements to Dam suspicious callers above all receiver ought to be allowed to not break any vital data to the offender next topic is configuration of viip adapters USB viip phone adapters enable USA to use any as ient telephone to position VIP calls they typically appear as if USB adapters these USB adapters have a typical standard Telephone Phone Port so that one will connect a daily connective once hooked up your phone operates as if connected to traditional public utility software controlled VIP applications soft phones soft phone's area unit phones that enable USA to create viip phone calls directly from a PC that has a web affiliation soft phones build decision with the assistance of a laptop telephone receiver and a sound card a

soft phone sort of a traditional phone just with the distinction that the affiliation is coming back from your laptop viip suppliers devel soft phones for free of charge in exchange of availing their service code controlled VIP application permits users to speak to others dedicated VIP phones exploitation these services at no further price a viip phone feels like a daily telephone an evid viip phone connects on an electronic network instead of a standard telephone line an evid VIP phone May match in two ways that a phone and a base station that connects to the net it should operate on a neighborhood wireless network dedicated viip phones need a supplier further as a service setup dedicated routers these routers in viip enable

user to attach standard phones to the net to position VIP calls wherever the router is connected to associate degree ADSL cable electronic equipment and permit users to connect a standard telephone viip routers have the extra practicality of associate degree science router as that permits USA to attach to the laptop further viip suppliers piece these routers underneath a particular setup for a charge these routers may also perform their functions freelance 27 minutes, 1 second of any PC or any code Wireless compatibility of viip with the wireless router mobile devices and smartphones may also use the viip system if you put in a wireless computer network you would like a applicable security like a firewall or

secret writing problems with wireless VIP VIP on computer network is deployed principally in company environments like in corporations instead of homes Wireless VIP poses issues of measurability for Enterprises as just in case with all wireless network quality of service is poor compared to wide networks setup price and maintain prices higher during a wireless VIP Network due to variety of excess points among the restricted space of network security threat is higher over a wireless VIP Network limitations of viip the popularity of viip can rely upon some key problems a number of these problems area unit a resultant of the very fact that science was designed for information packets whereas some problems stem from vendors that don't seem to be Meeting the Necessities to the

standards the key problems area unit mentioned as follows quality of service the design of science doesn't guarantee realtime transmission of voice packets as science was designed for information packets that guarantee error-free ordered delivery of information packets acceptableness of viip depends on delay that should not exceed a given threshold worth prioritization of packets means that giving voice packets high priority will guarantee smart quality of voice interoperability you need to Interchange the sign mechanism of pstn with viip sign mechanism if you wish to create viip common among net users a number of the suitable sign mechanisms area unit h. 3 to three standards sip protocols and mgcp every of those protocols integrates

information voice and video over constant wire security the question of security arises as a result of the backbone of uip is that the net that isn't a really secure medium there will be interception of calls due as fraud Security will be provided by exploit ation tunneling protocols like layer to tunneling the secret writing mechanism used in Secure sockets layer that is SSL however secret writing isn't wide out there for viip integration with public switch telephone Network that is pstn viip Works in Union with pstn and that they seem as one network to the users of the service in viip technology your telephone number has associate degree Science address therefore anytime a viip phone engages during a decision

its Science address is translated into the telephone number and two-handed over to the pstn network you would like to mix of each viip a pstn as a result of not everybody has switched to viip and their area unit are closer range of users still exploitation pstn scalability since calls over science have lower price and also the work on the rising the standard of voice further as transmission goes on there has been a high rate in VIP users the most obstacle lies in its measurability so viewers in today's lecture we discussed viip Protocols of viip Qs in viip and and security implementation on viip we also discussed the security measur and configuration of viip we will discuss the VIP protocols in detail in forthcoming lectures thanks

for watching this program thank you